10.15 The TEM mode that satisfies the conditions $\nabla\cdot\vec{E}_{TEM}=0$ and $\nabla\times\vec{E}_{TEM}=0$ can be found by solving the 2D Poisson equation, and the solution is the familiar $\vec{E}=\frac{\lambda}{2\pi\epsilon_0\rho}\hat{\rho}$, where $\lambda$ can be determined by $\int_a^b \vec{E}\cdot\hat{\rho}\,d\rho=\frac{\lambda}{2\pi\epsilon_0}\log\left(\frac{b}{a}\right)=V$, and $\lambda=2\pi\epsilon_0 V/\log(b/a)$. Applying the vector Smythe-Kirchhoff approximation, and assuming, without loss of generality, the polarization is $\vec{\epsilon}_1$, we have the incident field as

$$\vec{E}_i=\frac{\lambda}{2\pi\epsilon_0\rho}\vec{\epsilon}_1 e^{ikz}, \quad \text{and} \quad (\vec{n}\times\vec{E}_i)_{z=0}=\frac{\lambda}{2\pi\epsilon_0\rho}\vec{\epsilon}_2 .$$

Now, the diffracted field is

$$\vec{E}(\vec{x})=\frac{ie^{ikr}}{2\pi r}\vec{k}\times\int(\vec{n}\times\vec{E}_i)e^{-i\vec{k}\cdot\vec{x}'}da'=\frac{ie^{ikr}}{2\pi r}\frac{\lambda}{2\pi\epsilon_0}(\vec{k}\times\vec{\epsilon}_2)\int_a^b\rho\,d\rho\int_0^{2\pi}d\beta\,\frac{1}{\rho}e^{-ik\rho\sin\theta\cos(\phi-\beta)}$$

$$=\frac{ie^{ikr}}{r}\frac{\lambda}{2\pi\epsilon_0}(\vec{k}\times\vec{\epsilon}_2)\int_a^b J_0(k\rho\sin\theta)\,d\rho$$

If the polarization vector is given by $\vec{\epsilon}_i=\vec{\epsilon}_1\cos\alpha+\vec{\epsilon}_2\sin\alpha$, the final result will be

$$\vec{E}(\vec{x})=\frac{ie^{ikr}}{r}\frac{\lambda}{2\pi\epsilon_0}\left[\vec{k}\times(-\vec{\epsilon}_1\sin\alpha+\vec{\epsilon}_2\cos\alpha)\right]\int_a^b J_0(k\rho\sin\theta)\,d\rho$$

$$=\frac{ie^{ikr}}{r}\frac{\lambda k}{2\pi\epsilon_0}\left(\vec{\epsilon}_1\cos\theta\cos\alpha-\vec{\epsilon}_2\cos\theta\sin\alpha+\vec{\epsilon}_3\sin\theta\cos(\phi-\alpha)\right)\int_a^b J_0(k\rho\sin\theta)\,d\rho.$$

Let $I=\int_a^b J_0(k\rho\sin\theta)\,d\rho$, we have

$$\frac{dP}{d\Omega}=\left(\frac{V}{\log(b/a)}\right)^2\frac{k^2}{2Z_0}\left(\cos^2\theta+\sin^2\theta\cos^2(\phi-\alpha)\right)\left[\int_a^b J_0(k\rho\sin\theta)\,d\rho\right]^2$$